

PrepBench ADF course: second accuracy pass on the remaining material

2026-10-04 · Changed: `backend/app/content/packs/adf/v1.json` , `docs/research/adf-documentation-notes.md` , `docs/research/lessons/unit-01-loading-only-new-data.md` · Builds on the first audit report; first-pass corrections were not reopened.

A. Executive summary

Scenarios 5 to 18 cannot be audited: their full text does not exist anywhere in the accessible repository. I found 25 points in the rest of the uncovered material: **8 were correct and are kept, 2 were incorrect, 14 were incomplete, overbroad, context-dependent, misapplied or unverifiable** (qualified or removed), and **1 is a first-pass claim of mine that I am withdrawing in part**. Verification used 29 further Microsoft Learn pages, listed with URLs in section 31.5 of the research notes.

Correction to my first audit. I said the "5 to 7 minutes" data flow warm-up figure was no longer on any current Microsoft page, and changed the course on that basis. That was too strong. It is not on the overview or debug-mode pages, but Microsoft's "Delta lake ETL with data flows" and "Best practices for writing to the lake" tutorials still say "Data Flow clusters take 5-7 minutes to warm up". The performance guide says 3 to 5 minutes for a job cluster. The course now says "several minutes" and gives both figures with their sources. The first PDF and its summary are wrong on this one point; the repository text and notes are now corrected.

B. Scenarios 5 to 18: what exists

I searched the committed pack; the whole prototype tree (`units.ts` , `movingData.ts` , the prototype guide copy, lab components, the Gemini review files); the three archives (`prepbench-work-backup-2026-09-27.zip` , `prototypes.zip` , `final-review-bundle.zip`), unpacked and searched by content; the lessons folder; the implementation plans and gate reports; backend seeds and the frontend source. I searched for every scenario title and for `UNIT_5` to `UNIT_18` .

- Only scenarios **1 to 4** are fully written (`UNIT_1` in `units.ts` ; `UNIT_2` to `UNIT_4` in `movingData.ts`), identical to the pack. The archives hold older copies of the same four.
- `units.ts` states it in its header ("Scenarios 1 to 4 ... are written; the others are planned"), and the content-packs plan says "18 scenarios, 1-4 written".
- So, for each of scenarios 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17 and 18: **this scenario could not be audited because its full learner-facing content is not present in the accessible repository**. I did not invent or reconstruct any.
- What exists for each is a title, a one-line outcome, a source reference and a chapter link. I checked those 14 lines against the verified facts: all are technically valid. Two outcome lines were sharpened (scenario 7 in the first pass; scenario 13, "with the session log on", now). The full table is in notes section 31.1.
- Scenarios 1 to 4 were re-read only for statements that touch the uncovered material. One was wrong (below, F-19).

C. Changes made

Where	Original problem	Verified Microsoft behaviour	Change made	Source page	Severity
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Bad rows chapter	Fault tolerance "skips bad rows" with no limits stated; a duplicate key listed as a skip case; log presented as automatic.	Not applied to Synapse PolyBase loads, Redshift UNLOAD, stored-procedure SQL sinks, or Upsert into a SQL sink. Primary-key skips only for SQL Server, Azure SQL Database, Cosmos DB. Skipping and the session log are separate settings; the log is off by default.	Rewrote the block: three documented cases, the exclusions, skip-versus-log, the file-level options. Updated the recovery table, the security leak list, a diagnostic point and two new pitfalls.	copy-activity-fault-tolerance; copy-activity-log	High
Bad rows chapter, consistency	No mention of <code>Unsupported</code> or <code>NotVerified</code> , the unsupported connectors or staged copy.	Result values Verified / NotVerified / Unsupported; not supported for FTP, SFTP, HTTP, Snowflake, Office 365, Databricks Delta Lake or staged copy; block-level checksum, file-level MD5 only if the source carries it; tables: row count only.	Added all of it, kept the "counts, not correctness" distinction, and said "check what it actually said".	copy-activity-data-consistency	Medium
Bad rows chapter, type conversion	Correct, but the June 2020 date had no stated purpose.	The page still says "since late June 2020"; older activities don't show the options; code-created pipelines default to off; truncation default true.	Kept the date, explaining it only as the reason old and new pipelines differ. Added an interview takeaway.	copy-activity-schema-and-type-mapping	Low
Reconciliation chapter	"An If version of each" aggregate; "empty values"; no hash caveat; data flows assumed to read any source.	countDistinct and approxDistinctCount have no If variant; countAll includes NULLs. All named functions exist; no median or percentile function appears on the reference. Oracle, Teradata and Netezza are not data flow sources.	Fixed both statements; "no median" now dated; crc32 is a short checksum; hashes need identical types, formats and NULL handling; land non-data-flow sources with Copy first.	data-flow-aggregate-functions; data-flow-assert; data-flow-expressions-usage; data-flow-source	Medium
Transform chapter	Schema drift off was described as "the load fails".	Without Allow schema drift, columns outside the projection aren't picked up. Validate schema is the option that fails the flow. Drifted columns are text unless inferred; drift costs early binding.	Rewrote the block around the three real options and kept the decision framing. Folder partitioning now notes its small write cost. Debug start-up wording corrected (see A).	concepts-data-flow-schema-drift; data-flow-source; tutorial-data-flow-write-to-lake	High
Loading-only-new-data chapter; Unit 1 lesson	"Microsoft's tutorial and its template connect the watermark update to success."	Tutorial step 30 connects the Copy's Success output to the Stored Procedure. The template page lists the four activities but does not name the exit.	Both texts now cite the tutorial only and tell the learner to check the template. Added a labelled note that the three-pattern chooser comes from the 2020 SAP white paper and that current docs list more options.	tutorial-incremental-copy-portal; solution-template-delta-copy-with-control-table; tutorial-incremental-copy-overview	Medium

Migration chapter; diagnostic question	10 TB partitioning stated generally; example "took about 146 hours"; "ADF is built for"; DistCp described as "fastest".	The 10 TB advice is for native IR mode. All worked examples are explicitly hypothetical. Microsoft says "well-suited". DistCp is recommended; the SHIR only submits and monitors, so its size doesn't set throughput.	Scoped the advice, reworded the example to "would take" with a course note, added the two-week example, replaced "fastest". Diagnostic point scoped.	data-migration-guidance-hdfs-azure-storage; ...-netezza-...; ...-s3-...; ...-overview	Medium
Sensitive data chapter (new block and table row)	"Connect ADF to Purview to discover and classify sensitive data and trace its lineage."	ADF pushes runtime lineage for Copy, Data Flow and Execute SSIS only; Purview drops lineage for unsupported stores; query and stored-procedure sources unsupported; data flow shows source and sink only; classification comes from Purview scans. (Microsoft's ADF security page uses the same shorthand as the course did.)	Rewrote the row and added a "what Purview adds, and what it doesn't" block with an interview takeaway.	connect-data-factory-to-azure-purview; purview/data-map-lineage-azure-data-factory; secure-your-azure-data-factory	High
Fabric chapter; diagnostic question	"Dataflow Gen2: a different engine"; mount "generally available since May 2025".	Microsoft documents Spark-based mapping data flow transforms in Gen2 (preview). The GA dates are not on the pages I could reach.	Qualified the Gen2 cell and diagnostic point; rewrote the three upgrade paths from the page and removed the unverifiable date.	fabric upgrade-planning page; compare page	Medium
Performance and tuning chapters	Duration table presented without its assumption; "DIUs" as step 2; shuffle partitions without the skew caveat; drop-indexes advice unqualified.	Table values match. Microsoft calculates it as achievable with tuning and many concurrent copies. Start from default DIUs; dedicated SHIR machine. More shuffle partitions don't help skewed data. "Document per line" JSON uses little memory in data flows.	Added each qualification.	copy-activity-performance; ...-troubleshooting; concepts-data-flow-performance; concepts-integration-runtime-performance	Low
Scenario 4 (written)	The tumbling-window debrief said "Use UTC: hourly schedules don't shift for daylight saving, but daily ones do."	The daylight-saving rule is documented for schedule triggers, not tumbling windows.	Reworded: give the trigger UTC start times; the DST rule belongs to schedule triggers.	concepts-pipeline-execution-triggers; how-to-create-tumbling-window-trigger	Medium
What-it-is chapter; white papers	The 2018 white-paper journey read as current architecture.	Concept still current; product names have moved (SQL DW is now a Synapse dedicated SQL pool).	Added a historical-example note. Classification of all four papers is in notes section 31.3.	purview/data-map-lineage-azure-data-factory (naming)	Low

D. Checked and found correct (left alone)

- "By default the copy fails" on an incompatible row, and the three tabular skip cases (with the sink scope added).
- The session log's contents: skipped rows appear with their values (security point stands).
- Consistency verification for tables is a row-count check, and "verified" does not mean business-correct. Preserved.
- Type conversion defaults and the June 2020 wording.
- The error-handling table (Try-Catch, Do-If-Else, Do-If-Skip-Else) and the leaf rule, re-fetched.
- The duration table values, tuning order, copy stage guidance, cluster-size table, shuffle-partition figures, TTL behaviour, logging levels.
- Delta writing with Alter Row and key columns; "name file as column data" needing a single partition.
- Migration sizing (D32s_v3; two nodes on the HDFS and S3 pages, four on the Netezza page), and the 100 GB / two-hour split.
- The Purview Data Curator role.

E. Remaining risks

Area	Risk
Scenarios 5 to 18	Do not exist as text. When they are written, they should be audited against notes section 31 before being added to the pack. I did not write them.
Microsoft disagreeing with itself	Warm-up time (5-7, 3-5, "a few" minutes); SHIR nodes to start with (two vs four); Purview wording; a copy-pasted crc32 description; an S3 example typo. The course states ranges or the common ground.
Volatile figures	Duration table, shuffle-partition defaults, cluster sizes, 45-day history, limits. All come from pages that change.
Fabric	Gen2 mapping data flow transforms are preview; GA dates for mount and pipelines unverified and not in the course.
Preview and UI	Portal paths and option names (Validate schema, session log) were read from documentation, not observed in a live factory.
Not checked	Individual connector pages; the full text of the four white papers (claims checked via current pages); tumbling-window dependency page; Databricks Job, Execute Pipeline, Web, Webhook and Script activity pages; the prototype's older <code>guides/adf.ts</code> (not shipped).

F. Validation

Check	Result
JSON and structure	Parses. 21 chapters, 18 scenarios (4 written), 10 diagnostic questions, unchanged. Every block is heading/md/table only; every table row matches its header width; no bold in table cells; no TODO or placeholder text.
Cross-references	Every "chapter N" mention is exactly the set <code>test_content_packs.py</code> pins.
Backend tests	Full suite from a throwaway git worktree (the learner's app is running): 966 passed, 2 skipped, 0 failed (6 min 44 s). Includes <code>test_content_packs.py</code> and the real-database isolation check.
Frontend unit tests	Full run on its own, nothing else running: 99 files, 1,196 tests, all passed . The first audit's run had one timeout under concurrent load; this run had no such failure.
Not run	The Playwright browser suite (its guide test checks only the 21-chapter count and titles, unchanged).

Files	Three modified (the pack, the research notes, the Unit 1 lesson) and this report added. The earlier report PDF, the three pre-existing untracked files, and all application code are untouched. Nothing is committed.
Repo-wide sweep	Searched for the superseded claims ("If version of each", "built for moving", "took about 146", "tutorial and its template", "generally available since May"). None remain in learner-facing text; the only hits are the audit notes quoting the old wording.

G. Readiness

INTERVIEW-SAFE for the written chapters and the four written scenarios. The two statements I found to be plainly false (an If variant for every aggregate; schema drift off "fails the load") and the overbroad ones (Purview, Gen2, fault tolerance scope) are corrected and labelled. **The course as a whole cannot be rated higher**, because 14 of its 18 listed scenarios have no text, and several supporting facts depend on Microsoft pages that contradict each other. It is not production-reference quality.